System Verification and Validation Plan for SFWRENG 4G06 - Capstone Design Process

Team 17, DomainX

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Revision History

Table 1: Revision History

Date	Developer(s)	Change
October 17, 2025	Awurama, Fei	Initial v0 Draft
March 6, 2026	Fei	Update Unit Test section
March 7, 2026	Haniye	Updated Functional Requirement Plan Section
April 3, 2026	Haniye, Ghena	Final documentation update contained listed in this Github issue.

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1 Symbols, Abbreviations, and Acronyms

Table 2: Symbols, Abbreviations, and Acronyms

Symbol / Acronym	Description
API	Application Programming Interface – mechanism for data retrieval (e.g., GitHub API, PyPI API).
AHP	Analytic Hierarchy Process – method for pairwise comparison and ranking of libraries.
CSV	Comma-Separated Values – export format for datasets.
DB	Database – MySQL instance used for persistent data storage.
UI	User Interface – front-end component built with React.
PoC	Proof of Concept – initial demonstration validating workflow integration.
VnV	Verification and Validation – process of ensuring correctness and meeting stakeholder needs.
CI/CD	Continuous Integration / Continuous Deployment – automated testing and deployment pipeline used in GitHub Actions.

All additional terms conform to those defined in the SRS Glossary (Section 4.1).

This document outlines the Verification and Validation (V&V) strategy for the Domain Assessment Tool capstone project. It defines how the team will confirm that the implemented system satisfies its specified requirements, performs reliably, and aligns with the objectives stated in the Software Requirements Specification (SRS), Development Plan, and Hazard Analysis.

The V&V Plan provides a structured roadmap covering requirement reviews, system and nonfunctional testing, and traceability between test cases and requirements. It also establishes the methods, responsibilities, tools, and metrics that ensure all deliverables are verified for correctness and validated against user and research expectations.

2 General Information

2.1 Summary

The Domain Assessment Tool is a web-based application that automates evidence collection, analysis, and visualization for assessing open-source libraries of various domains.

The tool replaces spreadsheet-based scoring with a traceable, auditable system that integrates the following automated features:

- **Automated Data Collection:** retrieval of repository metrics (commits, issues, languages, and lines of code) through the GitHub API and PyPI metadata.
- **Interactive Data Table:** centralized, editable interface for entering and verifying measurements.
- **Automated Analytic Hierarchy Process (AHP):** automated computation of pairwise comparisons for reproducible rankings.
- **Visualization and Export:** generation of graphs and reports in PNG and LaTeX formats for research use.

The V&V Plan defines how the software's correctness, reliability, and usability will be verified and validated throughout its lifecycle.

2.2 Objectives

The objectives of this V&V Plan are to:

1. Build confidence in software correctness by verifying that each functional requirement in the SRS (e.g., automated AHP calculation, data retrieval, visualization, and multi-user collaboration) is fully implemented and traceable to its tests.

2. Ensure system reliability and data integrity, particularly for automated data collection and storage operations identified as high-risk in the Hazard Analysis.

3. Validate usability and accessibility through internal surveys and pilot-user feedback from the Research Sub-team and Dr. Smith.

4. Confirm performance and security non-functional requirements, such as load handling and access-control validation.

Out of Scope:

- Third-party API correctness (GitHub, PyPI); assumed verified by their respective providers.

- Formal proof techniques and hardware-level testing; outside academic scope.

- Multi-browser performance optimization beyond the core compatibility tests specified in the SRS (Chrome, Firefox, Safari, Edge).

2.3 Challenge Level and Extras

Challenge Level: Advanced. Per the Problem Statement, the project integrates a custom software tool that automates data gathering, analysis, and visualization for various domains.

Although the broader capstone includes a research study on software-quality assessment, the scope of this V&V Plan pertains only to the verification and validation of the software tool itself, not to the accompanying research methodology or paper.

Extras (Approved by Supervisor): Usability Testing and Peer Code Reviews, conducted through structured user surveys and GitHub-based peer inspection checklists as defined in the Development Plan workflow.

2.4 Relevant Documentation

Table 3: Relevant Documentation and Their Relevance to V&V Activities

Document	Relevance to V&V Activities
Software Requirements Specification (SRS)	Defines functional and non-functional requirements that form the basis for test derivation and traceability.
Development Plan	Describes test environments, toolchains (PyTest, coverage.py, GitHub Actions), and team responsibilities used for verification automation.
Hazard Analysis	Identifies potential failures (e.g., data loss, access control errors, API failures) that inform stress and reliability tests.
Problem Statement & Goals	Clarifies project scope, stakeholders, and intended outputs to align validation activities with research objectives.
Design Document (MG/MIS – future)	Will provide module interfaces and algorithms for unit test mapping in Section **??**.

3 Plan

This section defines the overall verification and validation (V&V) strategy for the Domain Assessment Tool.

It outlines the V&V team structure, review methods for each lifecycle artifact, verification tools, and validation activities that will be conducted to ensure functional and non-functional compliance with the SRS.

The plan follows both static reviews and dynamic execution-based testing, combining automated pipelines with peer-driven inspections.

3.1 Verification and Validation Team

Table 4: Verification and Validation Team Responsibilities

Team Member	Role	Verification and Validation Responsibilities
Awurama Konadu Nyarko	*Research Lead / Usability & Validation Analyst / Project Coordinator*	Coordinates all V&V activities, including scheduling artifact reviews and maintaining traceability between requirements, hazards, and test cases. Oversees usability validation and ensures survey design, accessibility, and user feedback are integrated into validation outcomes. Consolidates peer feedback and manages version control of testing artifacts.
Fei Xie	*Scrum Master / Developer / Automation Engineer*	Oversees the implementation and automation of unit and integration tests. Maintains the CI workflow via GitHub Actions, configures coverage reporting tools (pytest-cov, flake8), and ensures regression testing is triggered on each pull request. Facilitates sprint reviews and ensures verification tasks are completed within iteration schedules.
Haniye Hamidizadeh	*SRS Verification Lead / Backend Expert / Developer*	Ensures the correctness, consistency, and testability of requirements. Leads peer inspection of the SRS, verifies traceability between functional and nonfunctional requirements and their corresponding tests, and validates that hazard-related mitigations are represented in the final design. Supports backend API verification and database validation.

Team Member	Role	Verification and Validation Responsibilities
Ghena Hatoum	*QA Lead / Backend Expert / Developer*	Designs and executes the formal testing plan, maintains the defect log, and coordinates peer reviews of test cases and results. Responsible for system-level validation, including black-box testing and regression verification. Oversees backend integration testing, verifies data-storage reliability, and ensures acceptance criteria are met before supervisor review.
Dr. Spencer Smith	*Supervisor / External Reviewer*	Provides independent oversight of all verification artifacts (SRS, Design Document, and VnV Plan). Reviews and approves testing procedures for alignment with course expectations and ensures documentation quality meets 4G06 capstone standards.

3.2 SRS Verification

Verification of the Software Requirements Specification (SRS) ensures that all requirements are consistent, unambiguous, complete, and testable.

The team will use a combination of peer inspection, supervisor walk-through, and structured checklist verification to confirm requirement quality and traceability.

Each requirement identified in the SRS (e.g., SR-AC1, SR-INT2, SR-IM4) will be reviewed using a checklist based on the IEEE 29148 standard, evaluating criteria such as correctness, completeness, consistency, verifiability, and traceability.

During the scheduled peer inspection session, team members will mark any ambiguous or unverifiable requirements for revision through the GitHub issue tracker under the label "SRS Review."

Feedback from the assigned primary reviewer team will also be incorporated to ensure external validation of requirement clarity.

As detailed in Table 4, each team member and the supervisor has a de-

fined role in the verification process. Awurama Konadu Nyarko coordinates overall V&V scheduling and documentation; Haniye Hamidizadeh leads SRS verification and requirement traceability checks; Fei Xie manages automated testing workflows; Ghena Hatoum oversees system-level validation and defect tracking; and Dr. Spencer Smith provides supervisory review and approval. This structure ensures that all SRS verification activities—from peer inspection to supervisor walkthrough—are aligned with the team's assigned responsibilities and maintain traceability across all artifacts.

Following the peer inspection, a structured 30-minute walkthrough will be conducted with the supervisor.

In this meeting, the team will present the SRS structure, explain traceability between requirement identifiers and their planned test cases, and clarify any areas of uncertainty.

All decisions, clarifications, and follow-up actions from the session will be logged in the repository's issue tracker to maintain a transparent audit trail of SRS verification outcomes.

The following checklist can be used:

- ☐ Are all major functions required for the web application (interface, backend, database) covered?
- ☐ Are requirements written with consistent terminology and layout?
- ☐ Do any requirements conflict with another one?
- ☐ Are requirements unambiguous and verifiable, is the fit criterion understandable?
- ☐ Are assumptions and constraints about user and system behaviours clearly stated?
- ☐ Is there a process to track the status of SRS Verification changes, are issues related to thee SRS Verification traceable?

3.3 Design Verification

The design verification process ensures that the software architecture and module interfaces correctly realize all functional and non-functional requirements defined in the SRS.

A structured design review will be conducted after completion of the Module Guide (MG) and Module Interface Specification (MIS) documents.

The team will:

- Perform checklist-based inspections of class diagrams, API schemas, and data-flow diagrams to confirm interface correctness, modularity, and traceability to SRS requirements.
- Use the issue tracker to document any inconsistencies or missing data-flow connections.
- Review database and API integration points to verify that the design addresses identified hazards such as data loss and synchronization errors.
- Hold a design walkthrough with the supervisor, during which reviewers will ask task-based questions (e.g., "trace the path of a data update request") to confirm the design supports required behaviour.

The checklist will evaluate design completeness, consistency, interface clarity, and maintainability:

- ☐ Are features traceable to at least one requirement from the SRS?
- ☐ Do UML diagrams follow convention?
- ☐ Are all updates and features traceable through the issue tracker in team's github repository?
- ☐ Are inputs and outputs of the system components clearly identified and correct?

3.4 Verification and Validation Plan Verification

The V&V Plan itself will be verified to ensure internal consistency and feasibility.

Planned activities include:

- **Peer review session:** each team member will inspect the document against the V&V Plan template and rubric, checking for section completeness, alignment, and internal traceability.

- **Supervisor feedback:** Dr. Smith will review the plan draft and provide comments prior to submission.
- **TA feedback:** the course TA will review the document for clarity, completeness, and adherence to course expectations, providing early feedback for improvement before final submission.
- **Version control:** changes and responses will be tracked in GitHub using pull-request comments and a "VnV Plan Review" label.
- **Mutation check:** minor edits (for example, changing requirement identifiers) will be tested to confirm that traceability references remain correct.
- **Issue tracking and resolution:** the team will review and update open GitHub issues related to verification and validation activities on a weekly basis. Each issue will be categorized by type (documentation, requirements, testing, or design), assigned to the responsible member from Table 4, and tracked to closure. This process ensures continuous improvement and accountability throughout the project's lifecycle.

The following checklist can be used:

☐ Does VnV Plan verify all functional requirements are met?

☐ Does VnV Plan verify all non-functional requirements are met?

☐ Have all peer-review issues been addressed and closed?

☐ Have all members of the team peer-reviewed the document prior to approval?

☐ Did test cases detect mutations and give desired outputs?

☐ Did test cases expected output match actual output?

☐ Is there a process for documenting and resolving incorrectness?

☐ Is there a process to track the status of VnV Plan changes, are issues related to thee VnV plan traceable?

3.5 Implementation Verification

Implementation verification ensures that each software component is correctly realized and adheres to design and coding standards.

- **Static verification:** code reviews will be conducted for all pull requests. Linters (`flake8`, `black` for Python, `ESLint` for JavaScript) will enforce syntax and style compliance (see *Development Plan*, Section 2).
- **Dynamic verification:** unit and integration tests written in PyTest will confirm correct module behaviour and interface compatibility.
- **Continuous integration:** GitHub Actions will automatically execute all tests on commit and generate coverage reports (target $\geq 80\%$).
- **Regression testing:** test suites will rerun automatically whenever code is merged to detect unexpected changes in functionality.
- **Code walkthroughs:** conducted at major milestones to compare implementation decisions with design diagrams and hazard mitigations.

3.6 Automated Testing and Verification Tools

Automated verification will rely on the toolchain defined in the Development Plan.

Table 5: Automated Testing and Verification Tools

Tool / Framework	Purpose
PyTest + `coverage.py`	Unit and integration testing with code-coverage metrics.
GitHub Actions CI/CD	Executes automated tests and generates build reports for every push or pull request.
`flake8` / `black` / `ESLint`	Static analysis and style enforcement for Python and JavaScript.
Selenium or Playwright	Automated front-end UI testing of key user workflows (domain creation and visualization).
Postman or `pytest-requests`	API endpoint validation and response-code checking.

Coverage summaries and logs will be archived in `/test/reports/` within the repository.

3.7 Software Validation

Software validation confirms that the delivered Domain Assessment Tool satisfies stakeholder expectations and performs reliably under normal operating conditions.

Validation combined functional scenario testing, usability evaluation, and non-functional performance checks as described below.

3.7.1 End-to-End Validation

Execute representative workflows, from data collection through AHP ranking and visualization, to confirm system integrity. These tests were performed manually by the research subteam using real library data.

3.7.2 Usability Assessment

Conducted with internal research users and Ryan Fang (Robotics domain student) using short post-demo surveys (questions provided in Appendix 7.2). The success criterion was a mean rating of ≥ 4.0 (out of 5) across all survey questions.

3.7.3 Performance Validation

Performance validation ensured the system meets the speed and latency requirements defined in the SRS (PR-SL1: AHP ranking calculation must complete in less than ten seconds).

Method:

- **Tools:** pytest with timing measurements, browser developer tools for front-end performance, and custom timing scripts for AHP calculations.
- **Timing:** Performance tests were conducted at two points:
  - **After feature completion:** Basic response time checks for new endpoints to detect performance regressions early.
  - **Before major releases (Rev 0 and Rev 1):** Full performance benchmark suite executed on representative datasets (5, 10, 20, and 40 libraries).
- **Metrics Collected:**

- Dashboard load time (median and 95th percentile)
- AHP recomputation time
- Filter application latency
- Export generation time

Pass/Fail Criteria and Remediation Plan:

- **Pass:** All measured response times are below the SRS thresholds (AHP $\leq$ 10 seconds; other actions $\leq$ 5 seconds for interactive use).
- **Fail:** Any response time exceeds the defined threshold.
- **Remediation Actions if Fail:**
  1. Identify bottleneck using profiling tools (Django Debug Toolbar).
  2. Optimize database queries (add indexes, reduce N+1 queries).
  3. Implement caching for expensive calculations.
  4. Defer non-critical computations to background tasks (Celery).
  5. If threshold still not met after optimization, document the limitation and negotiate revised threshold with supervisor.

3.7.4 Security Validation

Attempt invalid inputs and authentication edge cases to ensure protection against hazards such as data leakage and unauthorized access. This includes SQL injection attempts (prevented by Django ORM), privilege escalation tests, and invalid login attempts.

Pass/Fail Criteria:

- **Pass:** All unauthorized actions return HTTP 403 Forbidden; audit logs capture all failed attempts; no data leakage occurs.
- **Fail:** Any unauthorized action succeeds or sensitive data is exposed.
- **Remediation:** Immediate patch of security vulnerability and re-validation before next release.

3.7.5 Stakeholder and User Validation

Regular validation was conducted through:

- **Weekly stakeholder meetings:** Dr. Spencer Smith reviewed implemented features and provided feedback on alignment with requirements.
- **User testing sessions:** The research subteam and Ryan Fang (Robotics domain student) used the tool and provided feedback on usability, missing features, and pain points. This feedback was tracked via GitHub issues and prioritized for subsequent sprints.

3.8 Timeline and Iterative Process

The V&V activities followed an iterative process aligned with the team's agile development methodology. The timeline for verification and validation was structured as follows:

Table 6: V&V Timeline and Iterative Process

Activity	**Frequency / Timing**
Build Feature	Continuous, each sprint (2-week cycles)
Unit Testing	Immediately after feature completion, automated via CI/CD (pytest)
Integration Testing	After each sprint, before merging to main branch
Stakeholder Meeting	Weekly with Dr. Spencer Smith to review progress and verify requirements
User Feedback Session	Conducted with research subteam and Ryan Fang (Robotics domain student)
Dev Team Meeting	Weekly to review progress, assign tasks, and adjust priorities
VnV Plan Update	After each feedback session, as needed
Regression Testing	Before each release (Rev 0, Rev 1)

Iterative Process Description:

1. **Build Feature:** The development team implemented a new feature according to the sprint plan.

2. **Unit Testing:** Automated unit tests were written using pytest and executed via GitHub Actions CI/CD pipeline. Features without passing unit tests could not be merged.

3. **Integration Testing:** After feature completion, integration tests verified that the new feature worked correctly with existing system components.

4. **Weekly Dev Team Meeting:** The team reviewed test results, identified failing tests, and prioritized fixes. Adjustments to the VnV plan were discussed based on observed outcomes.

5. **Weekly Stakeholder Meeting:** Dr. Smith reviewed progress, provided feedback on feature implementation and test coverage, and identified any gaps in requirements verification. All changes were tracked through GitHub issues.

6. **User Feedback Session:** The research subteam and Ryan Fang (a Master's student who will use the tool for the Robotics domain) used the tool and provided feedback on usability, correctness, and missing features. This feedback directly informed updates to both the feature backlog and the VnV plan.

7. **VnV Plan Update:** Based on feedback from testing, stakeholder review, and user sessions, the VnV plan was updated to reflect new test cases, modified success criteria, or changed priorities.

8. **Regression Testing:** Before major releases (Rev 0 and Rev 1), the full test suite was executed to ensure no existing functionality was broken.

This iterative approach ensured that verification and validation were continuous activities rather than end-of-cycle events, allowing for early detection of issues and incorporation of stakeholder feedback throughout development.

4 System Tests

Test IDs follow the pattern `T-XX-#` where `XX` denotes the subsystem (AC = Access Control, DM = Domain Management, VR = Visualization, NF = Non-Functional).

4.1 Tests for Functional Requirements

This section describes the dynamic verification activities that will confirm the implemented system meets all functional requirements in the SRS.

Functional tests are grouped by feature areas that reflect the system's major use cases. All tests will be executed after integration and will rely on automated CI pipelines (`pytest`, GitHub Actions) supplemented by manual UI verification for user-facing features.

Each test case references the corresponding **SRS requirement identifier (SR–xx)** to maintain full traceability (see Section **??**).

4.1.1 Area of Testing 1: User Management and Access Control

Objective: Verify that user account creation, login, role-based access control, and authentication operate correctly and protect system integrity.

Test ID	Description / Procedure	Expected Outcome	SRS Refs
T–AC–1	Submit valid login credentials.	User is authenticated and redirected to the dashboard.	**FR3**
T–AC–2	Login as a Contributor and attempt to edit metric values.	User can view and edit metric values through the interactive table.	**SR–AC2**
T–AC–3	Login as a Super Admin and access administrative functionality.	User can access user management features and manage accounts.	**SR–AC2**
T–AC–4	Attempt login with invalid credentials or unregistered email.	Access is denied; error message displayed; no session created.	**SR–AC5, UH–UP3, LF–AR6**
T–AC–5	Update user role (e.g., Contributor to Super Admin).	Role is updated successfully and permissions reflect the change.	**SR–AC4, PR–SC3, FR7**
T–AC–6	Authenticated user updates password through profile page.	Password is updated and required for future logins.	**SR–AC5, PR–SC1**
T–AC–7	Access system as a public or unauthorized user and attempt modification.	Published data is viewable but modification actions are restricted.	**SR–AC1**
T–AC–8	Super Admin creates user and triggers password setup email.	User receives email, sets password, and account becomes active.	**FR2**
T–AC–9	Attempt to access restricted functionality without sufficient privileges.	Restricted actions are hidden and direct access is blocked or redirected.	**SR–AC6, SR–IM1**
T–AC–10	Perform password reset using email token.	Reset link is valid for one use and password is updated securely.	**FR4**

Verification Approach:
White-box unit tests (`pytest`) will validate authentication and token logic.
Black-box UI tests (`Selenium`/`Playwright`) will simulate login, role-based access, and password workflows.
Manual security validation will include privilege-restriction checks and unauthorized access attempts, referencing hazards identified in the Hazard Analysis (e.g., improper access control, credential misuse).

4.1.2 Area of Testing 2: Domain and Data Management

Objective: Verify that domains, libraries, and metrics can be created, modified, and retrieved accurately, and that automated data collection and AHP ranking behave as specified.

Test ID	Description / Procedure	Expected Outcome	SRS Refs
T–DM–1	Create a new domain.	Domain is stored in database with unique identifier and displayed in UI.	**FR5, SR–INT1, SR–AC3**
T–DM–2	Add library using GitHub repository URL.	Repository metadata (commits, issues, LOC) is retrieved successfully via external tools.	**FR7, FR10, OE–IA1, SR–INT2, SR–INT3**
T–DM–3	Edit metric values for a library.	Updated values are stored and AHP ranking is recalculated accordingly.	**FR7, FR12, SR–INT4, SR–IM2**
T–DM–4	Simulate API failure during repository analysis.	System reports failure and preserves previously stored metric data.	**PR–RFT1, PR–RFT2, SR–INT5**
T–DM–5	Export domain data as Excel file.	Downloaded file contains complete and correct domain and metric data.	**FR14, SR–INT7, OE–PR4**
T–DM–6	Publish a completed domain.	Domain becomes publicly accessible for viewing and download.	**FR6, OE–PR3, FR14**

Table 8: Functional test cases for domain and data management.

Verification Approach:
Dynamic system tests will validate database persistence, API integration,

and data processing.
Automated UI tests will confirm front-end updates and form validation.
Failure scenarios (e.g., API timeout, missing data) will be tested to ensure robustness and data integrity, referencing the Hazard Analysis.
Export and publishing workflows will be verified through integration testing and file validation.

4.1.3 Area of Testing 3: Visualization and Reporting

Objective: Verify that visualization components and export functionalities correctly display and summarize analyzed data.

Table 9: Functional test cases for visualization and reporting.

Test ID	Description / Procedure	Expected Outcome	Control	SRS Refs
T–VR–1	Generate visualization for selected libraries and metrics.	Comparative graphs are displayed correctly and reflect stored values.	Auto	**FR13, FR12, SR–VIS1, SR–IM1**
T–VR–2	Modify selected metrics or libraries and regenerate graphs.	Graphs update dynamically to reflect new selections.	Auto	**FR13, SR–VIS2, PR–SL1**
T–VR–3	Download generated visualizations.	Visualization images are exported successfully (PNG).	Manual	**FR15, OE–PR4, SR–INT7**
T–VR–4	Resize browser or test on different screen resolutions.	Graphs remain readable and layout adapts correctly.	Manual	**LF–AR2, LF–SR2, UH–AR1**

Table 9 (continued)

Test ID	Description / Procedure	Expected Outcome	Control	SRS Refs
T–VR–5	Generate visualization with missing metric values.	System handles missing data without runtime errors.	Auto	**PR–RFT1, SR–INT5**
T–VR–6	Open Manage Metrics page.	All metrics and definitions are displayed correctly.	Manual	**FR11, UH–LR2, LF–AR3**

Verification Approach

- **Dynamic Tests:** Executed using `pytest` for backend validation and Playwright or Selenium for UI verification.
- **Static Verification:** Manual inspection will confirm readability, layout consistency, and labeling accuracy.
- **Cross-Validation:** Visualization outputs will be compared against stored metric values and AHP rankings.
- **Failure Handling:** System behavior under missing data and API failures will be validated against Hazard Analysis controls.

4.2 Tests for Nonfunctional Requirements

This section specifies execution-based experiments (with summaries or plots rather than simple pass/fail where appropriate) and structured static reviews for the Domain Assessment Tool. The scope aligns with the SRS's nonfunctional sections (Usability, Performance, Robustness/Fault-Tolerance, Maintainability, Security) and with hazards identified in the Hazard Analysis.

Note on accuracy: For this project, "accuracy" concerns tool computations and outputs (e.g., AHP ranking consistency), not machine-learning model accuracy (explicitly out of scope). Relative error will be reported against a manual baseline as per the fit criterion.

4.2.1 Performance & Scalability Experiments

Goal: Measure how quickly the tool responds as the number of libraries and metrics increases.

Method (Automated Test):

- Record the time to load dashboards, apply filters, recalculate AHP scores, and export results.
- Run tests on datasets of 5, 10, 20, and 40 libraries.
- Save the median and 95th-percentile response times and plot them against dataset size.

Expected Output: A table and graph showing how response time grows with dataset size; any slowdowns will be noted for optimization.

Traceability: PR-SL1 (Speed and Latency Requirement).

Development Plan Tools: Automated using `pytest` and GitHub Actions performance workflows.

4.2.2 Reliability and Fault Tolerance

Goal: Confirm the system remains stable, consistent, and capable of recovery when services fail or operations are interrupted.

Method (Automated + Manual Tests):

- Simulate API errors (HTTP 429, 5xx) and verify that the tool uses cached data and displays a warning instead of crashing.
- Interrupt a mid-update transaction to ensure rollback to the last valid state.
- Perform a backup-and-restore test to confirm recovery within one hour.
- Verify that no partial or corrupted records are written during failures.

Expected Output: Concise test log summarizing behaviour under each error type, rollback event, and restore cycle, confirming that data integrity was preserved.

Traceability: PR-RFT1–PR-RFT3, SR-INT5.

References: Hazard Analysis — data loss and API outage risks.

4.2.3 Usability Evaluation

Goal: Check that the interface is intuitive, accessible, and that charts and exports are clear and easy to interpret.

Method (Manual Test + Survey):

- Have a small group of testers perform common tasks (create domain, add libraries, apply filters, export data).
- Record time taken, errors encountered, and number of clicks required per task.
- Ask participants to complete a short usability survey (Appendix B) rating ease of use, clarity, and navigation flow on a 1–5 scale.
- A mean rating of ≥ 4.0 across questions will meet the success criterion.

Expected Output: Average task completion times, click counts, and survey results summarized in a bar chart (Appendix C).

Traceability: UH-EU1 (primary navigation ≤ 3 clicks), UH-AR1 (font scalability), LF-AR2 (responsiveness across screens), LF-SR2 (minimalist graph design), UH-UP3 (clear error messages).

4.2.4 Computational Accuracy

Goal: Ensure that automatically computed AHP ranking results and auto-filled repository metrics are correct within an acceptable tolerance.

Method (Automated + Manual Comparison):

- Prepare a small AHP matrix and compute expected pairwise weights manually.
- Run the same inputs through the tool and compare outputs.
- For automatically fetched repository data, manually cross-check key fields (creation date, last commit date) against GitHub.
- Calculate relative error (%) for each metric; flag any deviation $> 2\%$.

Expected Output: Table comparing manual vs. tool-generated values, showing relative-error percentages and confirming compliance with thresholds.

Traceability: PR-PA1 (auto-filled data accuracy) and FR12 (AHP ranking computation).

4.2.5 Security and Access Control

Goal: Validate that user data, credentials, and permissions are properly secured, and that unauthorized actions are detected and prevented.

Method (Static + Dynamic):

- Perform a code review of authentication modules to confirm proper password hashing (e.g., Django built-in hashing) and secure session-token handling.
- Attempt unauthorized actions (e.g., deleting data without sufficient privileges, accessing admin-only routes) and verify they are blocked (e.g., HTTP 401/403 responses) and do not modify system state.
- Review logging configuration to ensure security-relevant events (e.g., failed authentication attempts) are recorded with timestamps and associated user/context where applicable.
- Verify that secure cookie and HTTPS settings are enforced in production configuration.

Expected Output: Completed security-verification checklist, screenshots/logs of simulated unauthorized access attempts, and confirmation that restricted actions are blocked with appropriate responses.

Traceability: PR-SC1–PR-SC3, SR-AC6, SR-IM1, SR-AU1.
Also aligns with Hazard Analysis controls for unauthorized access and data integrity risks.

4.2.6 Maintainability

Goal: Ensure that the codebase remains clean, modular, and easy to update, with consistent style and adequate automated-test coverage.

Method (Static + Automated):

- Run Python linters and formatters (`flake8`, `black`) to check for violations of the team's coding standards.
- Generate test-coverage reports from CI pipeline runs and record the overall percentage (target $\geq 80\%$).
- Conduct a brief team code walkthrough to review function naming, modular structure, and documentation completeness.

- Log all findings and proposed improvements in the GitHub issue tracker.

Expected Output: Lint-violation report, CI-coverage summary, and meeting notes capturing reviewer feedback and assigned follow-ups.

Traceability: MS-MR1–MS-MR6 (maintainability standards: style compliance, modularity, testability, documentation). Supports the Development-Plan verification process for maintainability compliance.

Format and Reporting Notes

- Accuracy tests use relative-error (%) instead of pass/fail.
- Performance and usability tests produce tables and graphs (Appendix C).
- Static reviews (walkthroughs, checklists) record who attended and what was found.
- All outputs are stored in the project's GitHub repository under `/test/reports/`.

4.3 Traceability Between Test Cases and Requirements

Table 10: Traceability Between Test Cases and Requirements.

Test ID	Requirement IDs (SRS / NFR)	Requirement Category	Rationale / Coverage Summary
T–AC–1	SR–AC1, SR–AC2, FR2, FR3	Access Control	Verifies account creation and login success paths.
T–AC–2	SR–AC5, UH–UP3, LF–AR6	Access Control / Usability	Confirms invalid logins are rejected securely with user-friendly feedback.
T–AC–3	SR–AC4, PR–SC3, FR7	Access Control / Security	Validates correct role updates and permission-based UI behaviour.
T–AC–4	SR–AC6, SR–IM1, SR–AU1	Access / Integrity / Audit	Ensures unauthorized actions are blocked and properly logged.

Table 10 (continued)

Test ID	Requirement IDs (SRS / NFR)	Requirement Category	Rationale / Coverage Summary
T–AC–5	FR4, SR–AC5, PR–SC1	Security / Access Control	Verifies secure password reset and credential hashing.
T–DM–1	FR5, SR–INT1, SR–AC3	Domain Management	Confirms domain creation and persistence with access validation.
T–DM–2	FR7, FR10, OE–IA1, SR–INT2, SR–INT3	Data Collection / Integration	Validates GitHub API metadata retrieval and field accuracy.
T–DM–3	FR7, FR12, SR–INT4, SR–IM2	Integration / Computation	Confirms manual metric edits and AHP ranking updates.
T–DM–4	PR–RFT1, PR–RFT2, SR–INT5	Reliability / Integration	Tests fallback to cached data and recovery under API failure.
T–DM–5	FR14, SR–INT7, OE–PR4	Export / Data Integrity	Checks JSON/Excel export completeness and file integrity.
T–DM–6	FR9, SR–INT1, UH–EU2	Data Validation	Confirms bulk-upload schema enforcement and error reporting.
T–DM–7	FR6, OE–PR3, FR14	Publication / Access	Verifies publishing workflow for completed domains.
T–VR–1	FR13, FR12, SR–VIS1, SR–IM1	Visualization	Validates accuracy and labeling of dashboard graphs.
T–VR–2	FR13, SR–VIS2, PR–SL1	Visualization / Performance	Confirms dynamic filter response within latency threshold.
T–VR–3	FR15, OE–PR4, SR–INT7	Export / Visualization	Verifies exported charts match on-screen visuals.

Table 10 (continued)

Test ID	Requirement IDs (SRS / NFR)	Requirement Category	Rationale / Coverage Summary
T–VR–4	LF–AR2, LF–SR2, UH–AR1	Usability / Accessibility	Ensures UI responsiveness and legibility on all screens.
T–VR–5	PR–RFT1, SR–INT5	Reliability	Confirms system handles missing values gracefully.
T–VR–6	FR11, UH–LR2, LF–AR3	Usability / Information	Checks accurate metric definition display and tooltips.
T–NF1 (Performance & Scalability)	PR–SL1	Performance	Measures response time vs. dataset size for dashboards, AHP, and exports.
T–NF2 (Reliability)	PR–RFT1–PR–RFT3, SR–INT5	Reliability / Integration	Evaluates stability, rollback, and recovery from service errors.
T–NF3 (Usability)	UH–EU1, LF–AR2, UH–UP3, LF–SR2	Usability	Assesses task efficiency and average survey rating ≥ 4.0.
T–NF4 (Computational Accuracy)	PR–PA1, FR12	Accuracy / Computation	Verifies AHP and auto-filled metrics within $\pm 2\%$ error margin.
T–NF5 (Security)	PR–SC1–PR–SC3, SR–AC6, SR–IM1, SR–AU1	Security / Access Control	Confirms encryption, authorization, audit logging, and lockout enforcement.

Table 10 (continued)

Test ID	Requirement IDs (SRS / NFR)	Requirement Category	Rationale / Coverage Summary
T–NF6 (Maintainability)	MS–MR1–MS–MR6	Maintainability	Verifies linting, modular structure, and CI coverage metrics.

5 Unit Test Description

We use pytest to implement unit tests for the system. All tests are done automatically through github actions, unless otherwise stated. Unit tests will be located in the test subfolder of the project. More detailed reports of the unit tests can be found in the VnVReport under the Unit Test section.

5.1 Unit Testing Scope

The scope of the unit tests will only be focused on modules developed by the DomainX team, which are:

1. Domains Page Module
2. Data Edit Module
3. User Management Module
4. Automated Metrics Module
5. Comparison Module
6. Visualization Module
7. Database Persistence Module

The WSS lists out all the modules and their use, syntax, and semantics. Our testing philosophy is to consider the general use case of the user, with the most expected function. We will also include a few use cases where the user is expected to encounter errors.

5.2 Domains Page Module

Tests for this module concerns the creation of a new domain, updating domain information and associated admins as creators to the domain. The test should provide assurance that mapping between creators and domains are working correctly, and business restrictions for creating new domains.

1. `test_create_domain_success`: The base case, creating a new domain with domain name and description should be successful and the newly created domain should be stored in the database.

2. `test_create_domain_missing_domain_name/description`: We required domains to have both a name and description, if a domain is being created without one or the other, an error response should be returned and won't be created in the database.

3. `test_list/delete_domains`: Basic functionalities such as listing all the domains and deleting a specific domain is working. Also testing that when deleting a domain, the mapping between creators and domains are destroyed.

4. `test_update_domain_sets_creators`: The mapping between admin(creators) and domains is crucial so researchers can be associated with the data gathered. We test that this connection can be created successfully, and existing domains can edit this connection.

5.3 Data Edit Module

Tests for this module checks that data is being stored correctly in the backend and that our validation and filtering mechanisms are working correctly so that we catch incorrect payloads.

1. `test_list_libraries_returns_only_domain_libraries`: Test if the API endpoint `/api/libraries/by_domain/{domain_ID}/` returns only the libraries belonging to the specific domain. As different domains can contain the same library, this function validates that it will return the correct associated data.

2. `test_create_library_requires_domain`: API endpoint `/api/libraries/` to libraries can only create a new library if it is associated with a domain.

3. `test_create_library_invalid_domain`: If a request to create a library is sent, there should be validations in place that the associated domain actually exists.

4. `test_create_library_success_sets_pending_and_returns_task_id`: When a new library is created, it automatically start an analysis of automated metrics.

5. `test_create_library_enqueue_raises_marks_failed_and_task_id_none`: There should be proper user feedback for a created library that has a failed analysis.

6. `test_patch/put_library_success`: Updating single or multiple field related to the library should provide user feedback and changes should be made in the database.

7. `test_put_library_rejects_non_github_url`: The tool depends on retrieving automated metrics from github, we test if a new library creation will be rejected if provided github url isn't actually from github.

8. `test_delete_library_success/fail`: Test both cases of deleting a library successfully and when deletion fails, such as provided library does not exist.

9. `test_library_metric_value_update_requires_metrics_dict`: Test the API endpoint of updating the associated metric values for a library will validate the payload and provide user feedback if it doesn't.

10. `test_library_metric_value_update_creates_value_and_evidence`: Since we depend on external APIs to populate some of the metrics, we should have tests to ensure the proper filtering is in place. This test will check if given a payload, request to update an existing metric is successful, and request to update a nonexistent metric will be ignored and won't be created in the database.

11. `test_metric_value_bulk_update_rejects_non_list`: Test that the format of the payload to request bulk updates is correct. This ensures there are less issues downstream which can result in potential missed data.

12. `test_metric_value_bulk_update_success`: We also test the base case that the bulk update is triggered with the correct payload, and the expected metric values are updated.

5.4 User Management Module

Tests for this module validates that our business logic determining the amount of functionality a user can do within the system by their role in the system is working as expected.

1. `test_list_users_superadmin/user`: Only superadmin should have the ability to list all the users in the system. Admin and user roles, as well as unauthenticated users should get a forbidden response when calling the endpoint.

2. `test_superadmin/admin/user_update_role`: For security, only superadmin should be able to update other user's roles. Both admin and user shouldn't be able to, and receive forbidden responses if they call the endpoint.

3. `test_superadmin/admin/user_edit_user_profile`: As usual, superadmin have the permission to edit user information for all. Admin and users can only edit their own profile and receive forbidden responses if they call the endpoint to edit someone else's profile.

4. `test_superadmin/admin/user_edit_domain`: We don't want anyone editing a domain so editing domain is restricted to superadmins, who can edit all the domains, and admins, who can edit their own domain. Normal users will get a forbidden response.

5.5 Automated Metrics Module

Tests for this module ensure the extraction of the associated automated metrics from Github's API, git_stats, and scc are successfully, and correctly stored in the database. An example set of tests for the response from Github's API:

1. `test_extract_repo_info_valid_url`: Tests that we extract the repository name and owner from github's api correctly.

2. `test_extract_repo_info_strips_git_suffix`: Tests that we remove suffix from github's response.

We also test that during the analysis and processing of automated metrics, the system displays to the user the correct status of current operations. An example such as:

1. `test_analyze_repo_task_success_updates_metrics_and_marks_success`: User should have feedback when the automated metrics are all successfully extracted and stored in the database for the associated library.

2. `test_analyze_repo_task_sets_running_status_early`: We should inform the user that something is happening in the system, even if we haven't actively started extracting from the library.

3. `test_analyze_repo_task_exception_marks_failed_and_raises`: If there is a failure, user feedback should be given and errors should be handled smoothly.

5.6 Comparison Module

Tests for this module consists of testing if the comparison of libraries within a domain is working as expected. It includes testing the AHP calculations and using different weights for categories. We also test if the module can gracefully handle error and provide user feedback when problems occur.

1. `test_ahp_calculations_success`: The base case, when we use the AHP calculations it should process the passed in data in the manner we expect.

2. `test_ahp_calculations_not_found`: Calling the comparison endpoint with an invalid domain id should return an error response, instead of failing.

3. `test_ahp_calculations_with_weighted_categories`: Using AHP calculations and adjusting the weight of categories, we should see the expected results based on the rules set.

4. `test_domain_comparison_returns_metrics_and_rows_with_values_and_evidence`: We have a multitude of data gathered for each library in a domain. We should be able to retrieve all of them successfully with their corresponding evidence (for the automated ones) in order to be displayed.

5.7 Database Persistence Module

The tests for this are in relation to the persistence of our models in the database. They will include the validation and serialization of the data for the following database models:

1. users
2. domain
3. libraries
4. library metric values

6 References

1. **V&V Plan Template (McMaster Capstone, Dr. Spencer Smith).** Source template defining section structure, verification guidelines, and notes for functional/nonfunctional testing. Used as the basis for all section headings and content organization.

2. **Software Requirements Specification (SRS) – DomainX.** Source of all requirement identifiers (FR, LF, UH, PR, MS, SR). Referenced throughout Sections 3 and 4 for test case traceability.

3. **Development Plan – DomainX.** Referenced for verification workflow, CI/CD tools (`PyTest`, `GitHub Actions`), and automation approach used in performance and maintainability testing.

4. **Hazard Analysis – DomainX.** Used for risk-based test design in reliability and security sections (API outage, rollback, and data-loss scenarios).

5. **Problem Statement and Project Objectives – DomainX.** Referenced for fit criteria in nonfunctional testing (accuracy $\pm 2\%$, usability $\geq 4/5$ score threshold) and for defining project scope (tool computation accuracy $\neq$ ML model accuracy).

7 Appendix

7.1 Symbolic Parameters

The following symbolic parameters are referenced throughout this document and may be updated centrally for consistency across all tests and scripts. Values marked as *team targets* are not defined in the SRS but used for validation thresholds.

Symbolic Constant	Description	Value / Traceability
`MAX_LOGIN_ATTEMPTS`	Number of consecutive failed login attempts before account lockout.	**Team target: 5**. Linked to **PR–SC2** (lockout after failed attempts).
`AHP_ERROR_THRESHOLD`	Maximum acceptable relative error between manual and tool-calculated AHP scores.	**Team target: $\leq$ 2%**. Related to **FR12** (AHP computation).
`USABILITY_TARGET_SCORE`	Minimum average usability rating required for success.	**Team target: $\geq$ 4.0 / 5.0**. Based on **UH–EU1** and **UH–UP3**.
`PERF_TEST_DATASETS`	Dataset sizes used for performance benchmarking.	**[5, 10, 20, 40] libraries**. From **PR–SL1** (speed/latency).
`CI_COVERAGE_TARGET`	Minimum required automated test coverage in CI pipeline.	**Team target: $\geq$ 80%**. Supports **MS–MR3–MS–MR4**.

Table 11: Symbolic constants used in validation and verification tests. Values marked "team target" are not fixed by the SRS.

7.2 Usability Survey Questions?

These are the user tasks referenced in provided to the user. Each question is rated on a 1–5 Likert scale (1 = Very Easy, 5 = Very Difficult) based on the difficulty it is to complete.

1. Create an account and login.
2. Create a new Domain.
3. Add Metrics to the Domain you created.
4. Add two libraries to your domain.
5. Fill out the manually added metrics for the libraries you added.
6. Export the data you gathered as a CSV for your domain.
7. Visualize and export raw metric values.
8. Visualize and Export your AHP calculations.

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Lifelong Learning.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

For the case of the VnVPlan only **Awurama Nyarko** was in charge of this document and so will be the only reflection given.

1. What went well while writing this deliverable?

   The document came together clearly and consistently. The structure followed the template well, and referencing between sections was organized and traceable. As well as referencing SRS requirements.

2. What pain points did you experience during this deliverable, and how did you resolve them?

   Creating long tables for test cases and mapping all SRS requirements to verification activities was time-consuming and required careful formatting. Aligning the tables with LaTeX syntax and ensuring traceability across all sections was challenging. This was resolved by breaking the tables into smaller parts, rechecking references, and using consistent labels and formatting commands.

3. What knowledge and skills will the team collectively need to acquire to successfully complete the verification and validation of your project? Examples of possible knowledge and skills include dynamic testing

knowledge, static testing knowledge, specific tool usage, Valgrind etc. You should look to identify at least one item for each team member.

Although I was the only person who worked on this document, the team collectively needs to strengthen our knowledge of automated testing frameworks (e.g., PyTest) and LaTeX documentation for technical reporting. We also need to deepen our understanding of continuous integration workflows and test coverage tools to support comprehensive verification.

4. For each of the knowledge areas and skills identified in the previous question, what are at least two approaches to acquiring the knowledge or mastering the skill? Of the identified approaches, which will each team member pursue, and why did they make this choice?

   For automated testing, the team will use online tutorials, official PyTest documentation, and hands-on practice through the project's CI pipeline. For LaTeX and documentation skills, we will review McMaster's templates, share formatting tips, and collaborate through Overleaf to maintain consistency. These approaches were chosen because they combine practical application with easily accessible learning resources that directly support the project's verification and reporting needs.